to do it, not just logically but pedagogically as well. That way we don't fall into the trap of thinking that quantum mechanics is basically just classical mechanics with a couple of new gimmicks thrown in. By the way, quantum mechanics is technically much easier than classical mechanics.

The simplest classical system—the basic logical unit for computer science—is the two-state system. Sometimes it's called a *bit*. It can represent anything that has only two states: a coin that can show heads or tails, a switch that is on or off, or a tiny magnet that is constrained to point either north or south. As you might expect, especially if you studied the first lecture of *Volume I*, the theory of classical two-state systems is extremely simple—boring, in fact. In this volume, we're going to begin with the quantum version of the two-state system, called a *qubit*, which is far more interesting. To understand it, we will need a whole new way of thinking—a new foundation of logic.